INTELLIFY 4.0
Project Abstract Submission Format

A. Team Details

Team Name: **Tech Titans**

Problem Statement ID: ALPHA409

College / University Name: Marwadi University

Team Leader Name: Jay Ramani

Contact Number: 8732944163

Email Address: jay.ramani137353@marwadiuniversity.ac.in

Team Members

Member 1: Jay Ramani

Member 2: Yogesh Parmar

Member 3: Shreya Raj

B. Project Information

Project Title: Multi-Modal Disinformation Detection and Propagation Tracker

Theme Selected: Artificial Intelligence & Social Awareness

1. **Problem Statement:** The rapid spread of fake news, propaganda, deepfakes, and misleading content on digital platforms makes it difficult for users to identify trustworthy information. Most existing systems only classify content as fake or real without explaining the manipulation techniques used or how misinformation spreads.

2. **Proposed Solution:**

- ❖ Develop NetraX AI, a multi-modal misinformation analysis platform.
- ❖ Analyse textual and visual content using AI models.
- ❖ Generate a Credibility Score based on source reliability and content analysis.
- ❖ Detect propaganda techniques such as:

  - **i.** Fear Appeals
  - **ii.** Emotional Manipulation
  - **iii.** False Authority
  - **iv.** Exaggeration
  - **v.** Bandwagon Effect

- ❖ Verify image authenticity and detect manipulated media.
- ❖ Visualize misinformation propagation through interactive network graphs.
- ❖ Provide explainable AI reports with evidence-backed insights.

3. Innovation / Uniqueness of the Idea:

- ❖ Detects propaganda techniques instead of only classifying fake news.
- ❖ Predicts audience psychological impact (fear, urgency, trust bias, sharing behavior).
- ❖ Combines text analysis, image verification, and propagation tracking in a single platform.
- ❖ Uses Explainable AI to justify every prediction with evidence.

- ❖ Predicts misinformation spread and identifies high-risk amplification sources.
- ❖ Promotes media literacy by helping users understand how misinformation influences society..

4. Expected Impact / Practical Use:

- ❖ Helps users identify misleading information before sharing it.
- ❖ Improves media literacy and critical thinking.
- ❖ Supports journalists and fact-checkers in content verification.
- ❖ Assists educators in teaching responsible information consumption.
- ❖ Helps organizations monitor misinformation campaigns.
- ❖ Promotes a safer and more trustworthy digital ecosystem.

5. Technologies / Tools to be Used:

- ❖ **Programming Language**
  - Python (AI model development, data processing, backend logic)
- ❖ **Frontend**
  - React.js (Build interactive user interface)
  - Tailwind CSS (Create responsive and modern designs)
- ❖ **Backend**
  - FastAPI (Connect frontend with AI models and process requests)
- ❖ **Database**
  - MongoDB (Store articles, credibility scores, reports, and user data)
- ❖ **Natural Language Processing (NLP)**
  - BERT (Text understanding and misinformation detection)
  - RoBERTa (Advanced propaganda detection)
  - Hugging Face Transformers (Pre-trained AI models)
  - Sentiment Analysis Models (Detect emotional manipulation)
- ❖ **Computer Vision**

- CNN (Detect manipulated images)
- CLIP Models (Compare image and text consistency)
- Deepfake Detection Models (Identify AI-generated media)

❖ **Network Analysis**
- NetworkX (Track misinformation propagation)
- Graph Analytics (Identify spread patterns and high-risk nodes)

❖ **Visualization**
- D3.js (Create interactive propagation graphs)
- Interactive Dashboards (Display insights and AI reports)

❖ **Deployment**
- Render / Vercel (Host and deploy the application)